Physical Chemistry CUME, Mar. 31st, 2017

Quantum

Q1. In quantum mechanics, the momentum operator is $\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$. Assume a wave-function is prepared in a state described by $\phi = e^{ikx-i\omega t}$, (this is the equation for the light produced by a laser) if the momentum of this state is measured, what is(are) the possible outcome(s)? Please also provide the probability for each outcome to be observed.

Q2. State the three requirements for a well-behaved wave-function.

Q3, Draw schematically the ground state and first excited state of a one-dimensional particle in a box.

Q4. What is the average momentum of the ground state of particle in a box? (Although you probably won't need it, according to the Euler identity: $\cos(kx) = \frac{e^{ikx}+e^{-ikx}}{2}$)

Q5. Draw the molecular orbitals formed by the four C $2p_z$ for a cyclobutadiene (in the X-Y plane). Please rank the molecular orbitals in order of increasing energy. Your answer to Q3 may provide some guidance to this question.